Reprinted from the

TRANSACTIONS OF THE SILVER SOCIETY

Series VII · Measures and Standards
First Era 248 · Numbers 1 and 2



THE QUANTITY OF TIDINGS AND THE CAPACITY OF LINES

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KANAICH: PUBLICATIONS OFFICE OF THE SILVER SOCIETY
F.E. 248

Transactions VII (F.E. 248), 1–14 Numbers 1 and 2

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SERIES VII · MEASURES AND STANDARDS

THE QUANTITY OF TIDINGS AND THE CAPACITY OF LINES

By THORNE*

Received in the Month of Autumn Dusk, F.E. 247; printed in two parts, in Numbers 1 and 2 for F.E. 248.

SUMMARY. How many messages a line of the speaking network can carry in a second, and whether the false marks of a tide year can be avoided, have hitherto been settled only by experience. This paper makes of the tidings in a message a quantity, measuring how much the recipient does not know beforehand of which message he will receive, apart from what it means; under three natural requirements it can only be the negative of the sum of each probability times its logarithm, and its unit is taken to be the choice. It follows that a line has a capacity, with noise or without; that below the capacity, if the noise falls upon each mark independently, there are encodings that make false readings as rare as one pleases, and above it there are none; that a speaking line of breadth W under evenly confused noise has the capacity $W \log_2(1 + P/N)$; and that to hide a message from one who overhears the line, the key must hold at least as many tidings as the message. The watch-books of the Tallin relay station, the drafts of the Gazette of the League and the copies of the Great Library give the examples: the redundancy of the Common Tongue, and the capacity of the line from Tallin to Kanaich in ordinary and in tide months, in which important messages need not be sent three times.

1. Introduction

The watch-books of the Tallin relay station record one thing day by day: how many marks in every ten thousand were received falsely. What was sent, the books never record. Along one line go the Gazette of the League, the prices of the cities, the figures of joint reckoning, and the hours sent out by the Tower of All Things; the line treats them all alike, and the watchmen need not know what they say, so long as the marks the receiver takes are those the sender sent. To measure, then, how much a line carries and how much it gets wrong, what can be measured is not the meaning of the messages, but how much the recipient, before he receives, does not know of which message he will receive. This paper makes of that not-knowing a quantity.

For more than fifty years since the speaking network was built, the messages between city and city, the figures of joint reckoning, and speech

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itself have all gone by this one network. Sending tidings has always been instantaneous in the League, and the difficulties of learning have never lain there; but instantaneous is not cheap. A long line needs paired speaking crystals, and a relay needs cleft cyan agate, and both are dear, so that how much a line can carry in a second is a matter of account. There is also noise upon the line: the tide of mana disturbs the resonance of the crystals, and the receiver then reads falsely. In ordinary years the line from Tallin to Kanaich reads some five marks in ten thousand falsely; in the tide year 244 the ten-year tide rose in the three tide months of Reawakening, Summer Haze and Scrolls, and in those three months there were on the average a hundred and fifty false marks in every ten thousand [4]. The remedy upon the network is to send an important message three times and read it by the majority, and so important messages are carried at a third of what the line could do.

Behind these two matters lie several questions that others have already met. Brewer proved that if the resonance of a pair of crystals can follow only changes of less than W beats a second, the line can tell apart at most $2W$ marks a second [2]. Manning pointed out that how much a message carries should be measured by how many different messages might have been sent, taking the logarithm of their number, so that when two messages are joined, what they carry is the sum of the two [3]. But Manning's quantity treats every message as equally likely, and real messages are not so: in the Common Tongue some letters are common and some rare, and after certain letters certain others almost always follow. Nor did he reckon the effect of noise.

A system for sending tidings may be divided into five parts, as in Fig. 1: the source, which chooses the message to be sent; the sender, which encodes it into marks upon the line; the line, which carries the marks, and by which noise enters; the receiver, which decodes the marks it receives back into the message; and the recipient. Part I of this paper treats lines without noise, Part II lines with noise, Part III continuous tidings such as speech, and Part IV the overhearing of a line. To avoid confusion, the letters, figures and glyphs that a source chooses I call signs, and the brights and darks upon the line, marks.

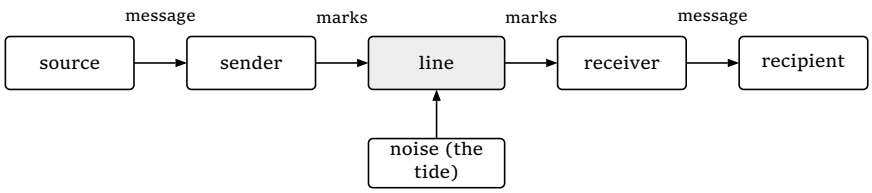


FIG. 1. The five parts of a system for sending tidings: the source chooses a message, the sender encodes it into marks, the marks pass along the line to the receiver, noise entering on the way, and the receiver decodes them into the message for the recipient

PART I. LINES WITHOUT NOISE

2. The capacity of a line

The marks a line can carry may differ in length: the old marks of the speaking network were a long beat, a short beat, and two pauses of different lengths [1]. In a time T, the strings of marks that can be laid out by the rules of the line are $N(T)$ in number. As T grows, their number grows roughly as an exponential of T, and I call

$$C = \lim_{T \to \infty} (\log_2 N(T))/T \tag{1}$$

the capacity of the line, in choices a second (for the choice, see Section 4). If all the marks are of one length, t seconds, and there are m kinds of them, then $C = (\log_2 m)/t$. The line from Tallin to Kanaich now uses two kinds of mark, bright and dark, of one length, at 2,400 marks a second, and its capacity is 2,400 choices a second. For the old marks of unequal length the capacity given by (1) is the root of an equation, which is given in the Appendix.

3. The source

A source chooses sign after sign. Which sign it chooses has a probability, and the probability of the next sign often depends on what the signs before it were. A source may therefore be regarded as a process that goes forward step by step by probability: at each step, each kind of sign has a probability of coming next, depending only on some number of the signs before. Messages in the Common Tongue are of this kind. Among the 1,000,000 letters that the Tallin relay station counted from the spoken drafts of the Gazette of the League, there are twenty-seven kinds, the space and the letters; the commonest, the space, is about a fifth of all, and the rarest is less than one in a thousand; in the Gazette, X and Z occur almost only in the names of cities and persons. The prices of the cities were counted apart, 200,000 letters, the figures in them being spelt out in letters in the drafts. I have used only public drafts, and no private messages.

I treat only sources of one kind: if the message taken is long enough, the frequencies not of single signs only but of every string of signs become steady, and they are the same from whatever part they are taken. I divided the drafts of the Gazette into ten parts, and the frequencies of the common single letters, pairs and triples differ from part to part by a few in a hundred at most. The prices and the Gazette are two kinds of writing, and taken together they would not be the same from whatever part they are taken, so that the two were estimated apart; the prices give some tenth of a choice less to the letter than the Gazette, and only the Gazette is used below.

4. The quantity of tidings

Let a source choose one of n possible messages, with probabilities $p_1, p_2, ..., p_n$. The quantity I want, written H, should satisfy three requirements:

(i) H should change continuously with the probabilities;

(ii) when all the messages are equally likely, H should be the greater, the more of them there are;

(iii) if one selection is broken into two steps, one after the other, H should be the H of the first step together with the H's of the second step, averaged by the probabilities of the first.

THEOREM 1. *The only quantity satisfying these three requirements is*

$$H = -K \sum_{i=1}^{n} p_i \log p_i \tag{2}$$

where K is a positive number that, with the base of the logarithm, fixes the unit.

The proof is given in the Appendix. I take logarithms to the base two and $K = 1$, and call the unit a choice: to decide one of two equally likely messages yields one choice. Manning's quantity is the particular case in which all the messages are equally likely, $H = \log_2 n$.

The quantity of tidings has several properties that will be used below. If every probability is nought or one, that is, if it is known beforehand which will be chosen, H is nought; for a given number of messages, H is greatest when they are equally likely. The tidings of two selections x and y taken together are no more than the sum of the tidings of each, and equal to it only when the two have nothing to do with one another. The tidings that remain in y when x is known, written $H_x(y)$, are on the average no more than those of y itself: on the average, knowing something can only lessen what is not known; the two are equal only when x and y have nothing to do with one another. For a selection between two possibilities with probabilities p and $1 - p$, the tidings change with p as in Fig. 2: one choice when p is one half, and near nought when p is near nought or one.

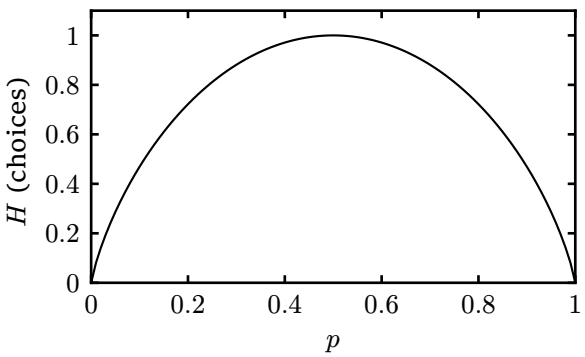


FIG. 2. The tidings (in choices) of a selection between two possibilities, one of which has probability p

5. The tidings and the redundancy of sources

The tidings of a source to the sign are the tidings it gives on the average to each sign, choosing sign after sign. Where the signs before affect the next, the tidings must be reckoned separately for each case of the signs before, and

the mean taken by their probabilities. In this manner I have estimated three sources, as in Table I.

Table I. The tidings to the sign of three sources (choices)

Source	*Estimated by*	*Tidings to the sign*	*Marks now used to the sign*	*Redundancy*
Common Tongue	all 27 equally likely	4.75	5	5%
	single letters	4.10	5	18%
	the letter before	3.38	5	32%
	the two letters before	3.02	5	40%
	the copyists' guessing (upper bound)	1.4	5	72%
Figures of joint reckoning	digits equally likely (upper limit)	3.32	4	17%
Glyphs of the code repository	frequency of glyphs	9.8	20	51%

The redundancy is one less the ratio of the tidings to the sign to the marks used, the present codes being taken for the measure. In the rows for the Common Tongue, by single letters a letter holds 4.10 choices, while the present code uses five marks for every letter, a redundancy of eighteen in a hundred; the further back one looks, the fewer the tidings and the greater the redundancy. The row of the copyists was made by twelve copyists of a copying-house in Copyists' Street, Chaen: they were given the first half of a number of the Gazette and asked to guess onward letter by letter, being told when they were wrong, and the number of guesses each letter needed was recorded; the twelve guessed two thousand four hundred letters in all. From the numbers of guesses an upper and a lower bound of the tidings to the letter can be reckoned; the upper is some 1.4 choices and the lower some 0.7, and the upper bounds of the twelve severally lie between 1.3 and 1.6. This manner of estimating is rough, and I take only its size from it. The figures of joint reckoning are now sent at four marks to the digit by the rules of the Seventh Division [6], after the old manner of recording a digit by the brightness and darkness of several runes that has come down from the star-reckoning arrays; if all digits were equally likely a digit would hold 3.32 choices, but real figures need not be so, and this is an upper limit. For the glyphs of the code repository, the Code Table of the Great Library fixes five decimal digits for every glyph [5], and in sending a glyph takes twenty marks; the copying office of the Great Library sent copies of the repository to the cities over the speaking network from 242 to 246, and counting the glyphs of 600,000 of them I find that a glyph holds only 9.8 choices.

6. Encoding

THEOREM 2. *Let a source give H choices to the sign, and a line have a capacity of C choices a second. Then there is an encoding by which the signs sent on the average in a second exceed $C/H - \varepsilon$, however small ε may be; and by no encoding can the signs sent on the average in a second exceed C/H.*

The heart of the proof is this. When a source sends a long string of signs, the longer the string, the nearer the logarithm of the probability of nearly every string that actually occurs, divided by the number N of signs in it, comes to $-H$; there are about 2^{NH} such strings, and the rest together have a probability as small as one pleases. If each of these 2^{NH} or so strings is given its own code upon the line, the time the codes take comes near to NH/C; the rest are given long codes of their own, and on the average they add almost nothing to the time. Conversely, a line can lay out only so many strings in a given time, and the source has at least so many to send, so that to send faster two strings would have to share one code.

In practice one may encode by halving: the kinds of sign are set in order of probability from greatest to least and cut into a former and a latter group whose probabilities are as nearly equal as may be; the codes of the former begin with bright and of the latter with dark; each group is cut again in the same way, until each holds but one kind. What is likely gets a short code; what is unlikely, a long one. No code so made is the beginning of another, so that the receiver, reading mark after mark, finds the letters apart of itself, without pauses between them. Both encoding and decoding can be written as formulae in Ashby's universal units [10], for every engine room to use. Coded in this way, the twenty-seven letters of the Common Tongue take on the average 4.14 marks to the letter, seventeen in a hundred fewer than the present five, and very little above the 4.10 choices estimated from single letters. If the code table is changed with the letter before, so that each letter is coded by the table belonging to the letter that precedes it, the twenty-seven tables together take on the average 3.45 marks to the letter; if neighbouring letters are coded two together instead, a letter needs at least 3.74 marks, the mean of 4.10 and 3.38, so that changing tables saves more than coding in pairs. The figures of joint reckoning are not halved but coded three digits together in a fixed code of ten marks, which takes 3.33 marks to the digit, seventeen in a hundred fewer than the present four. If the glyphs of the code repository are coded by their frequency, a glyph takes on the average 10.0 marks, half the present number. See Table II.

Table II. Marks used to the sign after encoding (reckoned for a line without false marks)

Source	*Marks used now*	*After encoding*	*Tidings to the sign*
Common Tongue, halving letter by letter	5	4.14	4.10
Common Tongue, table changed with the letter before	5	3.45	3.38
Figures of joint reckoning, three digits together (fixed)	4	3.33	3.32
Glyphs of the code repository, halving	20	10.0	9.8

The marks saved are what the line is paid for in vain. But these savings hold only upon a line without false marks. With codes of unequal length, one false mark sets the following letters wrongly apart until they fall into step again by chance; with tables changed by the letter before, one false letter sends the next to the wrong table. Upon a line with noise, therefore, such encodings must be used together with the checking codes of Part II, or a mark of alignment set in every so many letters.

PART II. LINES WITH NOISE

7. The noise of the tide

The line from Tallin to Kanaich passes through three relays of agate. The watch-books record day by day how many marks in every ten thousand were read falsely; I have taken 240 to 247, as in Table III. In ordinary years every month is near five in ten thousand, and the worst month not much above ten. In the tide year 244 the ten-year tide rose only in the three tide months of Reawakening, Summer Haze and Scrolls: those three months had a hundred and fifty false marks on the average, the worst of them, Summer Haze, two hundred and ten, while the other nine months stayed near five, and the mean of the year was forty-two. The years before and after the tide year were like ordinary years. Within a day too there is rise and fall, the false marks of the tide months gathering in the hours of high tide by the tide-tables.

Part of the noise of the tide can be reckoned beforehand: the rise and fall of the ten-year tide, the several constituents and the ebb and flow within a day can all be foretold by the tide-tables since Aldridge [7, 8]. What can be reckoned is not noise. In the tide months of 244 the relay station, upon the standby receiver of this line, moved the line that divides bright from dark with the height of the tide by the tide-tables, while the chief receiver, unmoved, took the same marks: the chief receiver read a hundred and fifty marks falsely in every ten thousand on the average of the three tide months, the standby seventy. The seventy that remain are what the tide-tables cannot

foretell: local surges of mana, which come mostly in tide years, and passing stirrings; these are the noise.

TABLE III. False marks in every ten thousand on the line from Tallin to Kanaich, and the capacity to the mark in choices

Year	*Mean*	*Worst month*	*Capacity by the mean*	*Capacity by the worst month*
240	4.6	9	0.994	0.990
241	5.1	11	0.994	0.988
242	4.9	10	0.994	0.989
243	5.4	12	0.993	0.987
244 (tide)	42	210	0.961	0.853
245	5.7	12	0.993	0.987
246	5.2	10	0.994	0.989
247	5.3	11	0.993	0.988

There is one more fault upon a line. Wards of law can stop the speaking crystals, and then the marks are not false but lost, and the receiver knows which marks it has not taken. The capacity to the mark of such a line is one less the probability of loss, much better than as many false marks; I do not treat it further.

8. The residual doubt and the capacity

Let the sender send x, and the receiver receive y. What the recipient still does not know of what was sent, after receiving y, is measured by $H_y(x)$, which I call the residual doubt. The tidings that actually arrive with every mark are

$$R = H(x) - H_y(x) \tag{3}$$

that is, the tidings sent less the residual doubt. One may not simply subtract the false marks: when a hundred and fifty marks in ten thousand are false in the tide months, the tidings that arrive are not less by a hundred and fifty choices, but by more than eleven hundred, for the recipient does not know which marks are false, and to point out which marks are false itself needs tidings. Suppose that beside the line there were another without noise, solely to tell the receiver which marks are false and how they should be put right; it may be proved that this line must send at least $H_y(x)$ choices to the mark, and need send no more. The residual doubt is thus exactly the tidings needed for correction.

The capacity of a line to the mark is the greatest value of (3) over all sources:

$$C_1 = \max\bigl(H(x) - H_y(x)\bigr) \tag{4}$$

If the line sends n marks a second, its capacity a second is $C = nC_1$. On a line without noise the residual doubt is nought, and (4) returns to the capacity of Section 2.

9. The fundamental theorem of a line with noise

Theorem 3. *Let the noise fall upon each mark independently of the others, with a fixed probability of a false reading; let the line have the capacity C a second, and a source give H choices a second. If H is less than C, there is an encoding by which the probability that a sign is read falsely after decoding at the receiver is as small as one pleases; if H is greater than C, no encoding can make the residual doubt less than $H - C$.*

The heart of the proof is as follows. Reckoning by the mark, take very long strings, N marks long. When the receiver has received a string, nearly all the strings sent that agree with it fall within some $2^{NH_y(x)}$ kinds. From the strings the line can take, choose at random, with the probabilities that give (3) its greatest value, 2^{NR} of them as codes, R being less than C_1. When the receiver receives a string, it looks for the code that agrees with it. A false reading comes when some other code also falls among the $2^{NH_y(x)}$ or so that agree with the string received; any code chosen at random falls there with a probability of about $2^{-N(H(x)-H_y(x))}$, so that the probability of a false reading is no more than about

$$2^{NR} \cdot 2^{NH_y(x)} \cdot 2^{-NH(x)} = 2^{-N(C_1-R)} \tag{5}$$

When R is less than C_1, this comes near nought as N grows. Since codes chosen at random are already so good on the average, there must be one set among them no worse than the average. The latter half follows from Section 8: the tidings that arrive are at most C a second, and so the excess $H - C$ can only remain as residual doubt. When H is exactly C, by leaving a very small part of the tidings unsent one can still make the probability that a sign is read falsely as small as one pleases, but in general not the probability that a whole passage is read without a single fault.

This theorem says that there is such an encoding, not that it is easy to find: the codes must be very long, and the receiver must compare many of them. But it shows something the watchmen had not thought of: false marks need not be avoided at the price of slowing to nothing; if only one sends more slowly than the capacity, false readings may be made as few as one pleases.

10. An example: a line of bright and dark

The line from Tallin to Kanaich sends two kinds of mark, bright and dark, and each mark is read falsely with probability p, bright being read as dark as often as dark as bright. By (4), the capacity of such a line to the mark is

$$C_1 = 1 + p \log_2 p + (1-p) \log_2 (1-p) \tag{6}$$

The last two columns of Table III are reckoned by (6). Before the dividing line is moved by the tide-tables, the readings at high tide lean as a whole to one side, and the false marks lean with them; then (6) is only a lower bound.

Both (6) and Theorem 3 want false marks that are independent and of a fixed probability; but the probability of the noise of the tide changes with the phase of the tide, and the false marks gather in the tide months and in the

hours of high tide. Reckoned by the mean probability of the year, the capacity to the mark in 244 is 0.961 choices; but a code that is to hold in the tide months must look to the capacity of the tide months: 0.888 choices on the average of the three, and only 0.853 in the worst. A code made only for the mean of the year will fail in the tide months; and to scatter the marks so that the gathered false marks are spread through the whole year, one would have to wait months to fill one group, which is of no use. Happily the phase of the tide can be reckoned. When both ends know the phase by the tide-tables, they can change the code and the rate with the phase, sending faster in ordinary months and slower in tide months. The tidings sent on the average to the mark through the year are then the mean of the capacities of the several hours, taken by the hours: in 244, 0.967 choices, more even than the 0.961 reckoned by the mean probability of the year; and when the dividing line is moved by the tide-tables, the capacity of the tide months rises to 0.940 choices, and the mean of the year to 0.980. This is just what Section 7 said: what can be reckoned is not noise. Even in the tide months, then, this line need send only some tenth less; with the dividing line moved, only six in a hundred less. Sending three times throws away two thirds.

Table IV compares several manners of proceeding in the tide months, all reckoned for false marks independent of one another. Sent three times and read by the majority, at a rate of one third, each mark keeps in the tide months a false reading of some six and seven tenths in ten thousand. Graves has given me a code of seven marks to the group [9]: four marks carry tidings and three check them, and if one mark in a group is false, the receiver can tell which and put it right; the rate is four sevenths, but each mark keeps in the tide months a false reading of some nineteen in ten thousand, more than when sent three times. Both fall far short of the capacity: by Theorem 3, if only the rate is below 0.888 in the tide months, or below 0.940 with the dividing line moved, the false readings left may be made as few as one pleases, though the codes must be far longer. If false marks gather, both manners do much worse; and sending three times is better done with some hours between than one after another. Fig. 3 shows how the capacity to the mark changes with the probability of a false mark, and where the several manners stand.

Table IV. Several manners of proceeding on the line from Tallin to Kanaich in the tide months of 244 (false marks independent of one another)

Manner	*Rate (tidings to the mark)*	*False readings left to the mark, line unmoved (in ten thousand)*	*False readings left to the mark, line moved by the tide-tables (in ten thousand)*
Sent as usual, unchecked	1	150	70
Sent three times, read by majority	0.33	6.7	1.5
Graves's groups of seven marks	0.57	19	4.3
By Theorem 3, at a rate below the capacity	below 0.888; moved, below 0.940	as few as one pleases	as few as one pleases

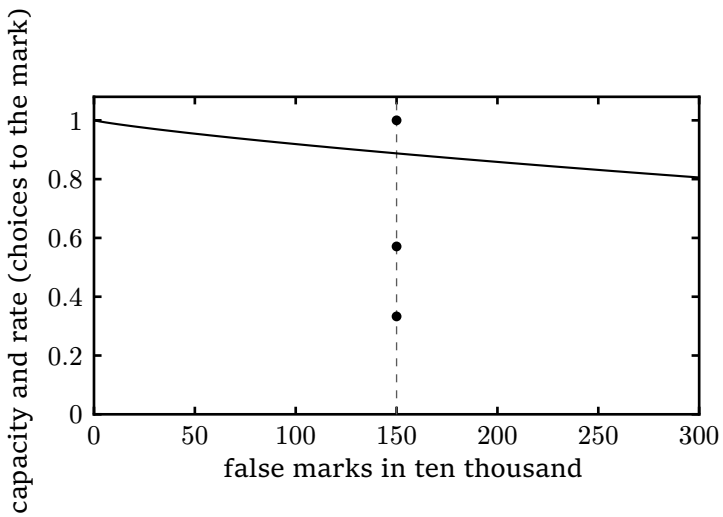


Fig. 3. The capacity to the mark (choices) of a line of bright and dark, against the false marks in every ten thousand; the three dots stand upon the upright broken line of a hundred and fifty, the mean of the tide months, and give the rates of the three manners of Table IV: sent as usual, groups of seven, and three times

PART III. SPEECH

11. Breadth and sampling

Speech is continuous: what is sent is a quivering that changes without pause, not mark after mark. The resonance of a pair of crystals follows only changes of less than W beats a second, and W I call the breadth of the line. Brewer's result [2] may be carried a step further: if a stretch of quivering contains only changes of less than W beats a second, it is fixed by its values at $2W$ evenly spaced instants in each second, and to know these values is to know

every instant of the stretch. A stretch of speech T seconds long upon a line of breadth W therefore has about $2WT$ numbers that may take their values independently; I say about, because a stretch that contains only changes within the breadth and is also exactly T seconds long cannot be made, and the longer T, the nearer the truth. The capacity of the line may then be reckoned after Parts I and II. How many tidings a continuous source itself needs is not a finite number: a number taking continuous values needs choices without end to be fixed, and only when it is settled how much difference may be allowed can it be said how many tidings the source needs; this I do not treat.

12. The capacity for speech

By evenly confused noise I mean this: the noise is added to the quivering sent, and has nothing to do with it; it is equally strong at all speeds within the breadth; and at every sample its value is spread according to the common spread of the classical study of errors, the samples having nothing to do with one another.

THEOREM 4. *If a line has the breadth W, the tidings reach the receiver with a mean power of no more than P, and the noise, of mean power N, is evenly confused, the capacity of the line is*

$$C = W \log_2(1 + P/N) \tag{7}$$

choices a second.

The heart of the proof: each of the $2WT$ numbers arrives mixed with noise, whose spread is fixed by N; the different values sent that the receiver can tell apart are about $\sqrt{(P+N)/N}$, and the $2WT$ numbers together about $((P+N)/N)^{WT}$; the logarithm divided by T is (7). Among noises that are added to the quivering sent and have nothing to do with it, of a given mean power, evenly confused noise is the hardest to meet, so that under other such noises the capacity is not less than (7). If the tide disturbs the very strength of the resonance of the crystals, that is not added noise, and (7) is not necessarily a lower bound; this I do not treat.

The ratio of the powers is found thus: the power of the stirring received at one end is measured while a given power is put in at the other, and again while nothing is sent; the second taken from the first is the power of the tidings received, which is set against the second. The line from Tallin to Kanaich has a breadth of some 1,200 beats a second; in ordinary years the ratio is some twenty-one, and in the tide months, with the dividing line moved by the tide-tables, some twelve. On this line a dark mark carries no power and a bright one carries it, the dividing line lying midway; under evenly confused noise such marks read falsely at these ratios with just the probabilities of Section 7, some five in ten thousand and seventy. By (7) the capacity in ordinary years is about 5,400 choices a second, and in the tide months about 4,450. The line now sends 2,400 marks a second, bright or dark, so that even with no false mark it sends only 2,400 choices, rather more than four tenths of its capacity in ordinary years. With several grades of brightness instead of

two kinds the same line could send far more: about $\sqrt{1 + P/N}$ grades can be told apart at each sample, some four or five in ordinary years and three or four in the tide months; how many are fitting, I do not treat here.

Where the noise is stronger at some speeds of quivering than at others, the sender's power should go rather to the speeds where it is weak; how the constituents of the tide fall upon the several speeds, I leave for another time.

PART IV. OVERHEARING

13. What the overhearer gains

Messages upon the speaking network pass through the hands of every relay along the line, and they are not hard to overhear. The present manner is for the two parties to agree beforehand upon a key, and to veil the message by the key before sending it. If one who overhears takes down the veiled message, and does not know the key, how much can he gain?

Theorem 5. *If one who overhears is to gain from the veiled message no tidings at all of the message itself, the tidings of the key must be at least as many as those of the message.*

Proof. Let the overhearer take down e; what he does not know of the message m is $H_e(m)$. For him to gain nothing, $H_e(m) = H(m)$. But when e and the key k are known, m is known, so that $H_e(m) \le H_e(k) \le H(k)$; together, $H(k) \ge H(m)$. And it can be done: let the key be as long as the message, each mark of it equally likely bright or dark, independent of the others and of the message, and used but once; and let the veiling add it mark by mark, like marks giving dark and unlike giving bright. Then what is taken down has nothing to do with the message.

The keys now in use are mostly phrases of ten letters or so, used again and again, holding no more than some fifteen choices, far fewer than the messages they veil, and the overhearer can in principle gain something; the greater the redundancy of the Common Tongue, the shorter the message he needs to take down. When he has taken down only a short piece, he has little to go on, and the redundancy of a letter, reckoned by the two letters before, is some 1.7 choices; in a long number of the Gazette, by the copyists' upper bound, some 3.4. With a key of fifteen choices, by the time some ten letters have been taken down only one possibility remains in principle for the message; this is only a matter of size. It supposes besides that the veiling is near to random; and it says only that the overhearer can, not that it is easy for him: actually to reckon it out depends on how simple or intricate the veiling is.

14. Conclusion

This paper has made of the tidings in a message a quantity, with the choice for its unit, and has drawn from it the capacity of a line, with noise or without, for marks and for speech. The conclusions have a common sense: how much a line can carry has a definite bound; in principle, if one sends more slowly than the bound, noise need not be feared; and the redundancy of a source

may be saved, or kept for checking, according to the encoding. The noise of the tide changes with the phase of the tide, and the phase can be reckoned, so that the capacity too is to be reckoned and used by the phase.

Several things are not done here: how good long codes are to be built, beyond Graves's example; how many tidings clear hearing needs, since speech must be heard rather than reproduced; the capacity under noise that is not even; and how hard a veiling is in fact to break.

I thank Graves and two old colleagues of Fenn for reading the first draft; the watchmen of the Tallin relay station for the watch-books and the trial of moving the dividing line; the copying office of the Great Library for its copies; and a copying-house of Copyists' Street, Chaen, for the trials of guessing.

Appendix. Proof of Theorem 1; the capacity of marks of unequal length

Take first the case where all are equally likely, and write $A(n) = H(1/n, ..., 1/n)$. When there is but one, it is known beforehand, and breaking it into two steps by (iii) gives $A(1) = 0$. To choose one of s^m equally likely messages may be broken into m steps, choosing one of s at each; by (iii), $A(s^m) = mA(s)$, and likewise $A(t^k) = kA(t)$. For any s and t not less than two and very large k, take m so that $s^m \le t^k < s^{m+1}$; by (ii), $mA(s) \le kA(t) < (m+1)A(s)$; and taking logarithms, $m \log s \le k \log t < (m+1) \log s$. The larger k is taken, the nearer $A(t)/A(s)$ comes to $\log t / \log s$, so that $A(t) = K \log t$; and since by (ii) $A(2) > A(1) = 0$, K is positive.

Then the general case. If every probability is rational, $p_i = n_i / \sum n_j$, the selection of one of $\sum n_j$ equally likely messages may be broken into first choosing i and then choosing one of the n_i in the i-th group; by (iii), $K \log \sum n_j = H(p_1, ..., p_n) + K \sum p_i \log n_i$, so that $H = -K \sum p_i \log p_i$. When the probabilities are not rational, the limit is taken by (i).

For the old marks of unequal length, $t_1, ..., t_m$ seconds long, any of which may follow any other, the C of (1) is the real root of $\sum_{i=1}^{m} 2^{-Ct_i} = 1$; the old marks had no rule on which may follow which, and a line with such rules has a smaller capacity, found from an equation of a determinant.

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